KAIZEN MATHS

Lesson starter

Hyperbolic Functions

Definitions and Identities

Name _____

S1

Write each hyperbolic function in exponential form.

Write sinh(3x) in exponential form.

S2

Find each exact value using the exponential definitions.

Find the exact value of sinh(ln 5).

S3

Prove each identity from definitions or known identities.

Prove that $\cosh x + \sinh x = e^x$.

S4

Simplify each expression using the cosh x + sinh x and cosh x - sinh x exponential identities.

Simplify 3cosh x - 3sinh x.

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ALG-6FDD-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Hyperbolic Functions

Definitions and Identities

Name

E1

Write each hyperbolic function in exponential form.

Write cosh(x) in exponential form.

E2

Find each exact value using the exponential definitions.

Find the exact value of tanh(ln 6).

E3

Prove each identity from definitions or known identities.

Prove that cosh x - sinh x = e^{-x}.

KAIZEN MATHS

Exit ticket

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Definitions and Identities

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E1

Write each hyperbolic function in exponential form.

Write cosh(x) in exponential form.

E2

Find each exact value using the exponential definitions.

Find the exact value of tanh(ln 6).

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Prove each identity from definitions or known identities.

Prove that cosh x - sinh x = e^{-x}.

ALG-6FDD-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Hyperbolic Functions

Definitions and Identities

Name

E1

Write each hyperbolic function in exponential form.

Write cosh(x) in exponential form.

E2

Find each exact value using the exponential definitions.

Find the exact value of tanh(ln 6).

E3

Prove each identity from definitions or known identities.

Prove that cosh x - sinh x = e^{-x}.

ALG-6FDD-L1 · E

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Hyperbolic Functions

Definitions and Identities

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E1

Write each hyperbolic function in exponential form.

Write cosh(x) in exponential form.

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Find each exact value using the exponential definitions.

Find the exact value of tanh(ln 6).

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Prove each identity from definitions or known identities.

Prove that cosh x - sinh x = e^{-x}.

ALG-6FDD-L1 · E

Show your working

Teacher answers & working

Hyperbolic Functions · Definitions and Identities

ALG-6FDD-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Write each hyperbolic function in exponential form.

Write sinh(3x) in exponential form.

Answer:

$\frac{e^{3x}-e^{-(3x)}}{2}$

Working:

Use the definition sinh u = $\frac{e^{u}-e^{-u}}{2}$.

Here u = 3x.

sinh(3x) = $\frac{e^{3x}-e^{-(3x)}}{2}$.

S2

Find each exact value using the exponential definitions.

Find the exact value of sinh(ln 5).

Answer:

$\frac{12}{5}$

Working:

Use $e^{\ln 5} = 5$.

Also $e^{-\ln 5} = \frac{1}{5}$.

sinh(ln 5) = $\frac{e^{ln5}-e^{-ln5}}{2} = \frac{5-\frac{1}{5}}{2}$.

Simplify to get $\frac{12}{5}$.

S3

Prove each identity from definitions or known identities.

Prove that cosh x + sinh x = e^x.

Answer:

cosh x + sinh x = e^x

Working:

Substitute the exponential definitions.

cosh x + sinh x = $\frac{e^{x}+e^{-x}}{2} + \frac{e^{x}-e^{-x}}{2}$.

The e^{-x} terms cancel.

cosh x + sinh x = $\frac{2e^{x}}{2} = e^x$.

Teacher answers & working

Hyperbolic Functions · Definitions and Identities

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S4

Simplify each expression using the cosh x + sinh x and cosh x - sinh x exponential identities.

Simplify 3cosh x - 3sinh x.

Answer:

$3e^{-x}$

Working:

Use cosh x - sinh x = e^{-x}.

3cosh x - 3sinh x = 3(cosh x - sinh x).

Therefore the expression simplifies to $3e^{-x}$.

E1

Write each hyperbolic function in exponential form.

Write cosh(x) in exponential form.

Answer:

$\frac{e^{x}+e^{-(x)}}{2}$

Working:

Use the definition cosh u = $\frac{e^{u}+e^{-u}}{2}$.

Here u = x.

cosh(x) = $\frac{e^{x}+e^{-(x)}}{2}$.

E2

Find each exact value using the exponential definitions.

Find the exact value of tanh(ln 6).

Answer:

$\frac{35}{37}$

Working:

Use $e^{\ln 6} = 6$.

Also $e^{-\ln 6} = \frac{1}{6}$.

tanh(ln 6) = $\frac{e^{ln6}-e^{-ln6}}{e^{ln6}+e^{-ln6}} = \frac{6-\frac{1}{6}}{6+\frac{1}{6}}$.

Simplify to get $\frac{35}{37}$.

Teacher answers & working

Hyperbolic Functions · Definitions and Identities

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E3

Prove each identity from definitions or known identities.

Prove that cosh x - sinh x = e^{-x}.

Answer:

cosh x - sinh x = e^{-x}

Working:

Substitute the exponential definitions.

cosh x - sinh x = $\frac{e^{x}+e^{-x}}{2}$ - $\frac{e^{x}-e^{-x}}{2}$.

The e^{x} terms cancel.

cosh x - sinh x = $\frac{2e^{-x}}{2}$ = e^{-x}.